SkyPass: Weather-Aware Satellite Pass Scheduling for Small Ground Stations and the Decision-Dependent Value of Cloud Forecasts

Rohith Prem S[1*] and R. Selvakumar[2]

[1*]Department of Computer Science and Engineering, Saveetha Engineering College, Chennai, 602105, Tamil Nadu, India.
[2]Department of Artificial Intelligence and Machine Learning, Saveetha Engineering College, Chennai, 602105, Tamil Nadu, India.

*Corresponding author(s). E-mail(s): rohithprem91@gmail.com;
Contributing authors: selvakumarr@saveetha.ac.in;

Abstract

Observation planning at a small ground station is split across unconnected tools: one predicts passes, another evaluates optical visibility, cloud is checked by hand, and overlapping passes are reconciled by the operator. This work examines whether an integrated, weather-aware planner increases the number of passes observed under clear sky, and under which operating conditions. The planner, SkyPass, carries a catalogue of orbital elements through SGP4 propagation, pass extraction by coarse bracketing and bisection, visibility scoring, cloud-forecast calibration against reanalysis, and exact conflict resolution, posed as weighted interval scheduling under a nightly budget and solved exactly by a dynamic program whose running time grows only linearly with the number of candidate passes and with the nightly observation limit. Forecast skill is verified against ERA5 reanalysis at seven stations and scheduling value at six, retrospectively over 60 days. Pass extraction agrees with an independent implementation to 0.055 s while using 23.7× fewer propagator evaluations than a dense scan, and the dynamic program matched exhaustive search on all 2000 randomised instances. A station required to observe every night gained nothing from the forecast, whereas one free to select its observing nights raised realised yield by 19.0% (expected value) and 37.6% (clear-sky count). Calibration raised the gain in every regime; uncalibrated, the forecast was worse than ignoring weather under a nightly quota. The value of a forecast therefore depends on the decision it informs: weather-aware planning pays when a station may choose its observing nights, and pays most with a calibrated forecast.

Keywords: satellite pass prediction, SGP4, ground station scheduling, interval scheduling, forecast calibration, optical observation

1 Introduction

Small ground stations, such as university laboratories, amateur-radio clubs and individual optical observers, face a recurring planning problem. Several hundred satellites are potentially observable and generate thousands of passes in a week, of which only a few dozen merit the operator's time, and the station attends to one object at a time.

Existing tools address only parts of this. Pass predictors compute rise and set times from two-line elements (TLEs) and the SGP4 propagator [1, 2], some adding an illumination test so that optical observers are not shown daylight passes. A few display a weather indication beside a pass, but to our knowledge none let the forecast *rank* passes or drive a scheduling decision, and none resolve overlapping passes.

Weather is the more consequential omission. At the site used as a running example (Chennai, India, 12.92° N), mean observed cloud over optically viable passes was 88%, under which most geometrically favourable passes are not observable, yet a predictor ranking by culmination elevation recommends an overcast night. The scheduling omission is less apparent: once several satellites are tracked their passes overlap and a single-aperture station has to choose. The literature treats that from the operator side of the link, as spacecraft tasking and range scheduling solved with mathematical programs, metaheuristics or learning methods [3–5], carrying constraints a small station does not have.

This paper shows that the small-station problem admits a substantially simpler exact formulation, and that the treatment of weather is what improves realised outcomes. The contributions are: an integrated pipeline accounting for the passes removed by each physical constraint; a validated pass extractor; exact capacity-constrained conflict resolution requiring neither a solver nor a metaheuristic; a weather claim verified against ERA5 reanalysis [6]; and two negative results, namely that weather-awareness yields no benefit under a fixed nightly quota and that the uncalibrated forecast degrades performance there.

2 Related Work

SGP4 and its two-line element sets remain the standard basis for rapid position prediction [1, 2], with accuracy limited by the elements rather than the propagator: errors grow with element-set age, measured against on-board GPS [7]. Whether a satellite is *seen* depends on illumination and photometry, treated with a conical Earth-shadow model [1] and a diffuse-sphere phase function [8].

Satellite range scheduling is an oversubscribed combinatorial problem [3], usually approached with multi-objective optimisation, metaheuristics or reinforcement learning [4]. Earth observation scheduling is a large adjacent literature [5], in which cloud cover appears as an uncertainty hedged during imaging tasking [9], although always on the *operator* side. Where the structure is single-machine interval selection,

exact polynomial algorithms exist: weighted interval scheduling is a textbook dynamic program [10].

Professional observatories schedule around weather with queue schedulers that replan as conditions change [11], but assume an institutional software stack and on-site sky monitoring. Forecast verification and forecast value are mature fields [12], and statistical post-processing is essential before a forecast drives a decision [13]. To our knowledge that machinery has not been applied to observer-side pass planning.

3 System Design

SkyPass is a five-stage pipeline (Fig. 1). Each stage is a pure function of the preceding one, so any stage may be ablated independently, which is the mechanism by which the weather claim is evaluated.

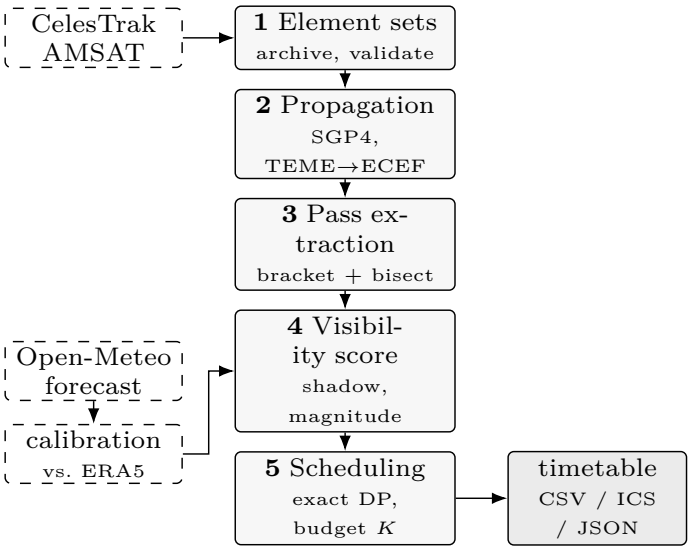


Fig. 1 SkyPass pipeline. The cloud forecast is calibrated against reanalysis before it is allowed to influence the score; the scheduler then resolves conflicts exactly under the station's nightly observing budget.

3.1 Element sets, propagation and pass extraction

Element sets are fetched from CelesTrak [14], checksum-validated, deduplicated per NORAD identifier keeping the newest epoch, and *archived under the date of retrieval*, since CelesTrak serves only current elements and without a local archive a published prediction cannot be repeated. Sets older than $\tau_{\max}$ are rejected. Positions come from the reference SGP4 implementation; the TEME vector is rotated to ECEF by Greenwich mean sidereal time and converted to topocentric azimuth, elevation and range

on the WGS-84 ellipsoid. Polar motion is neglected, being far under the SGP4 error budget, and refraction is reported but not applied to the horizon mask, so results do not depend on assumed surface conditions.

Because cost is dominated by propagator calls, pass extraction minimises them. Elevation is sampled on a coarse grid to *bracket* horizon crossings; since elevation within a pass is smooth and unimodal, a sign change of $\varepsilon(t) - \varepsilon_{\text{mask}}$ between adjacent samples brackets exactly one crossing. Brackets are refined by bisection to 0.1 s and culmination by golden-section search. The coarse step is $P/240$ bounded to 5 s to 30 s, so short-period orbits, whose passes are shortest, are sampled most finely.

3.2 Visibility scoring

Each pass receives a score $S \in [0, 1]$ combining a geometric quality term with multiplicative gates:

$$S = \underbrace{(w_e f_e + w_d f_d)}_{\text{geometry}} \cdot f_m \cdot f_s, \tag{1}$$

where $f_e = \min(\varepsilon_{\max}/60^\circ, 1)$ and $f_d = \min(T/480\,\text{s}, 1)$ saturate culmination elevation and duration, f_m is a photometric factor and f_s a sky-condition factor.

The multiplicative form is deliberate: an eclipsed pass or one in a bright sky scores zero irrespective of its geometry, and expected cloud scales the score down in proportion, which an additive score cannot represent. For optical work the object has to be sunlit and the observer in darkness; Earth shadow uses the dual-cone construction [1], and apparent magnitude follows the standard diffuse-sphere relation [8]

$$m = m_{\text{std}} + 5 \log_{10} \frac{d}{1000\,\text{km}} - 2.5 \log_{10} F(\psi), \tag{2}$$

with $F(\psi) \propto (\pi - \psi)\cos\psi + \sin\psi$ normalised to $F(0) = 1$. In radio mode the illumination and photometric terms are dropped.

3.3 Weather integration and calibration

The sky factor is $f_s = 1 - \hat{c}$, where $\hat{c}$ is the expected cloud fraction at culmination, interpolated linearly between hourly samples. Crucially $\hat{c}$ is *not* the raw forecast, which is heavily over-confident as a predictor. We fit

$$\hat{c} = \mathbb{E}[c \mid c_f] = a + b\, c_f \tag{3}$$

by least squares on a window that *ends before* the period being planned, restricted to local night hours, and $\hat{c}$ is clipped to $[0, 1]$. The slope b quantifies the signal: $b \approx 1$ means the forecast can be used directly, $b \approx 0$ that the calibrated value falls back to site climatology. For the threshold value model the planner instead weights each pass by the calibrated probability of clear sky, $P(c \le \theta \mid c_f)$, estimated in ten forecast bins on the same window and shrunk toward the site's base rate with 20 pseudo-observations, so each value model is paired with the planner that maximises its expectation.

3.4 Conflict resolution

A single-aperture station observes one object at a time and needs a setup gap g between targets, so choosing the best mutually compatible subset of passes is weighted interval scheduling. Writing $v_j = S_j \pi_j$ for the value of pass j (score times priority), indexing passes $1, \ldots, n$ by finish time and letting $p(j)$ be the largest $i < j$ compatible with j (0 if none), the textbook recurrence [10] gives the exact optimum in $O(n \log n)$.

That alone is insufficient: unconstrained, a station observes every candidate and weather-awareness has nothing to select. Real stations are limited by operator and instrument time, so a capacity budget of K observations per night adds a cardinality dimension:

$$\begin{aligned} \mathrm{OPT}[j][k] = \max\big(&\mathrm{OPT}[j-1][k], \\ &v_j + \mathrm{OPT}[p(j)][k-1]\big), \end{aligned} \tag{4}$$

with $\mathrm{OPT}[0][k] = \mathrm{OPT}[j][0] = 0$. This is exact. The table has one entry for each of the n candidate passes and each count up to the nightly limit K, and each entry is filled in constant time, so after the sort the cost is $O(nK)$: doubling either the candidates or the limit doubles the work. Observing nights are disjoint, so applying the budget per night and solving each independently stays exact. Nights are defined in local solar time, so an evening and the small hours that follow form one session.

4 Experimental Setup

The catalogue comprises 635 unique objects from CelesTrak GP groups, checksum-validated and deduplicated, observed from seven stations spanning distinct cloud climatologies: Chennai and Bengaluru (monsoon), Leh (high-altitude arid), Tucson (desert), Singapore (equatorial), and Reykjavík and Svalbard (maritime polar). Unless stated otherwise, the mask is 10°, the dark-sky threshold −6° and the setup gap 5 min. Timing is checked against a brute-force 1 s dense scan and against Skyfield [15], which shares only the reference SGP4 core with SkyPass. All runs are single-threaded under Python 3.13.5 with `sgp4` 2.27.

All three weather products come from Open-Meteo [16]: the operational forecast, archived past forecasts at lead times of 1–7 days, and ERA5 reanalysis [6] as verification truth. One methodological point silently invalidates the evaluation if overlooked: the archive endpoint has to be pinned to the ERA5 model, since at its default it returns a best-match series which, for recent dates, *is* the archived forecast, so verifying against it compares a forecast with a copy of itself and reports a spurious skill of 1.0.

A generative AI assistant (Claude, Anthropic) was used to help edit and restructure the manuscript text and to write the scripts that build it; the authors reviewed all such output and take responsibility for the content.

5 Results

5.1 Pass-prediction accuracy and cost

Table 1 compares SkyPass against both references over 635 objects and 1761 matched passes.

Table 1 Pass-timing agreement over the full catalogue (635 objects, 1761 matched passes, 24 h horizon). Values in seconds.

	vs. 1 s dense scan		vs. Skyfield	
Quantity	mean	p95	mean	p95
$\lvert\Delta t_{\mathrm{AOS}}\rvert$	0.504	0.954	0.060	0.157
$\lvert\Delta t_{\mathrm{TCA}}\rvert$	0.248	0.476	0.055	0.110
$\lvert\Delta t_{\mathrm{LOS}}\rvert$	0.502	0.956	0.059	0.156
$\lvert\Delta \varepsilon_{\max}\rvert$ (deg)	0.0008	0.0022	0.0034	0.0102

Against Skyfield, mean agreement is 0.060 s on acquisition and 0.055 s on culmination. The apparent 0.504 s "deviation" from the dense scan is not SkyPass error: a 1 s grid locates a crossing only to within its own step, so it lags the true crossing by a mean of 0.5 s, and the refined estimator is therefore more accurate than the brute-force reference it is conventionally compared against. The cost difference is large: 2 311 148 propagator evaluations against 54 864 000, a factor of 23.7. Detection recall is 99.78%, the residual being very short grazing transits that peak barely above the mask.

5.2 Element-set limits on prediction accuracy

Sub-second numerical agreement is meaningful only if the elements support that precision. The median element-set age in the operational catalogue was 0.42 days (90.2% under one day). CelesTrak and AMSAT sets differ by a mean of 0.041 s in culmination over 308 matched passes, but by a median of 0.00 s: the providers largely redistribute the same upstream sets, so this is not an independent check. Re-predicting the same passes from progressively older sets of 60 low-Earth objects gives 33 223 aged comparisons. Within a day of epoch the elements contribute 0.05 s, the same order as the 0.055 s numerical agreement of Section 5.1, so neither term dominates; degradation is then sharply super-linear and heavy-tailed, the mean reaching 4.3 s (median 0.70 s) at 7–14 days and 75.3 s (median 13.69 s) beyond a month, so a 5–7 day staleness cutoff suits timing-critical work and daily element refresh is the largest single lever on prediction accuracy.

5.3 Skill of the cloud forecast

Weather-aware planning is justified only if the forecast is informative at the lead times over which an observer plans. Archived forecasts were verified against ERA5 at all seven stations over 60 days, scoring the operational decision of whether an hour is

clear enough to observe at a 30% cloud threshold. Because accuracy misleads when clear hours are rare, we report the Heidke skill score, a chance-corrected measure in wide use for categorical verification [17]. Skill decays from HSS 0.347 at one day to 0.289 at three and 0.114 at seven, beating persistence (0.206) early but no longer by five to seven days, so a weather-aware planner is best run on a rolling short horizon. Skill is also strongly site-dependent, the spread between stations being comparable to the decay with lead time and lowest at the two monsoon stations: the expected benefit is a local property rather than a universal constant.

5.4 Value of weather-aware scheduling

Five planners select from an identical candidate set, drawn from 300 objects, at each of six stations over 60 days (Svalbard has no observable passes), and every timetable is scored against ERA5. Since the valuation of a clouded observation is a modelling choice, we report both an *expected value* model, $\sum_j v_j(1-c_j)$, and a *threshold* model, $\sum_j v_j \mathbb{1}[c_j \le \theta]$ with $\theta = 30\%$.

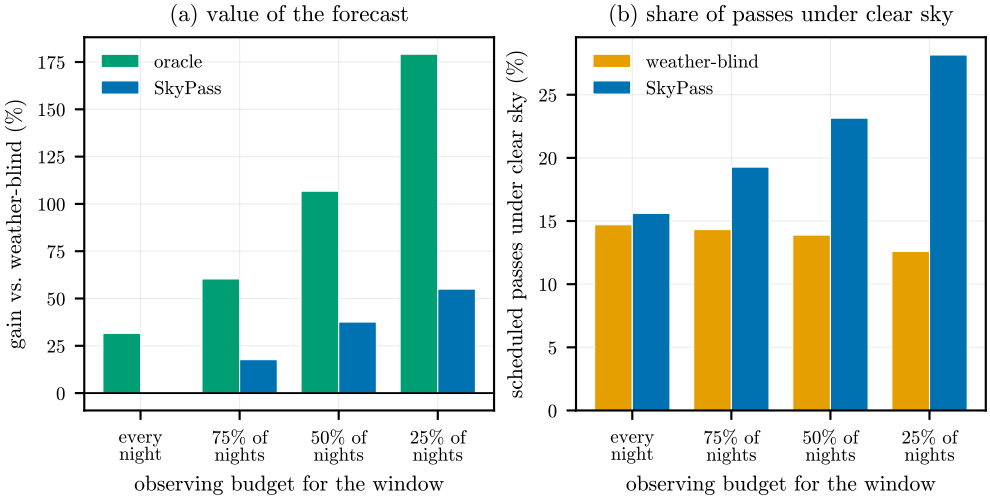


Fig. 2 Weather-awareness is worth nothing to a station obliged to observe every night, and a great deal to one that may choose its nights. (a) Gain over the weather-blind exact planner under the same budget, threshold value model; the oracle plans with the observed cloud and bounds what any forecast could achieve. (b) The share of scheduled passes that fell under clear sky.

Under a fixed nightly quota the forecast yields no benefit: −1.5% and −0.2% under the two models, against oracle headroom of 12.0% and 31.6%, and the station-bootstrap intervals of both gains include zero (−2.7 to 2.0% and −11.6 to 3.5%). It is not uninformative, since forecast and observed cloud correlate at $r = 0.41$ at the candidate-pass timestamps and the clearest-forecast decile does find clearer skies, reducing observed cloud by 0.209. The limitation is that a quota permits only *reordering within a night*, which two measured properties defeat: cloud is autocorrelated

over the hours a session spans, the between-night standard deviation being 2.2× the within-night value, while the ninetieth-percentile pass carries 1.8× the mean weather-free value. The score being a product, selecting for clear sky surrenders base value, an unfavourable exchange when pass quality is the more variable factor.

Given the same cap plus a window budget, so the station may *skip* a night, the result reverses (Fig. 2): which nights to use is a day-scale decision, the scale at which the forecast retains skill. We report the central of the three budget levels as the headline; Fig. 2 shows all four regimes. Observing on 50% of nights (90 observations), realised yield improves by 19.0% and 37.6% (95% station-bootstrap intervals 9.2–24.9% and 3.1–80.4%), recovering 41% and 35% of the attainable oracle gains; clear-sky passes rise from 13.9% to 23.1%, and a quarter-of-nights budget raises the threshold-model gain to 55.0% (24.7–144.5%).

Calibration raises the gain in every regime. Under the quota, supplying the model cloud field straight to the score costs −9.2% relative to disregarding weather; under the half budget the raw field still helps (8.0% and 29.9%) but less than the calibrated one. Fitted slopes lie between 0.17 and 0.61 and intercepts between 0.07 and 0.70, so clear forecasts are followed by more cloud than predicted and cloudy ones by less. Calibration (3) shrinks the forecast toward climatology in proportion to its demonstrated reliability.

Repeating the evaluation with epoch-correct geometry, that is, with the element set current on each night, raises the threshold-model gain under the quota from −0.2% to 9.2% and at a quarter budget from 55.0% to 75.9%, but lowers it at the half budget from 37.6% to 33.2%, the expected-value gain there being unchanged at 19.0%. Back-propagated geometry therefore does not inflate the result systematically, although it is not uniformly conservative.

5.5 Conflict resolution in practice

On 2000 randomised instances small enough to enumerate exhaustively, the dynamic program matched the optimum in 100.00% of cases. Greedy selection *by the score itself* attains 99.95% on average, so the dynamic program buys little in the average case; its advantage is being exact at negligible cost with no adverse case. The rules an operator applies fare worse, optimising a proxy rather than the objective: selection by highest culmination attains 95.39% on average, is optimal in only 62.7% of instances and falls to 9.4% at worst. An order-based genetic algorithm of the type common in the satellite-tasking literature (population 60, 120 generations) reaches 99.67% of the optimum but needs roughly 38 922× more time, a ratio that depends on that search budget; once the observer-side problem is recognised as interval scheduling, an exact polynomial solution is available and a metaheuristic is unnecessary. Scheduling is not the bottleneck in any case, the dynamic program handling 100 000 candidates in 285.3 ms against the seconds to minutes propagation needs.

5.6 End-to-end behaviour

The funnel is steep and physically interpretable (Fig. 3): at the demonstration site 12 449 geometric passes reduce to 8093 sunlit, 1297 in darkness, 367 bright enough to

detect, 87 above the score floor once forecast cloud is weighted in, and 26 scheduled, where a weather-blind tool would present the earlier, far larger counts. Per-station runs also check the physics: Svalbard (78.2° N) returns the most geometric and sunlit passes of any station yet *zero* observable ones, the Sun never falling below the −6° threshold in the planned week (29 August to 5 September).

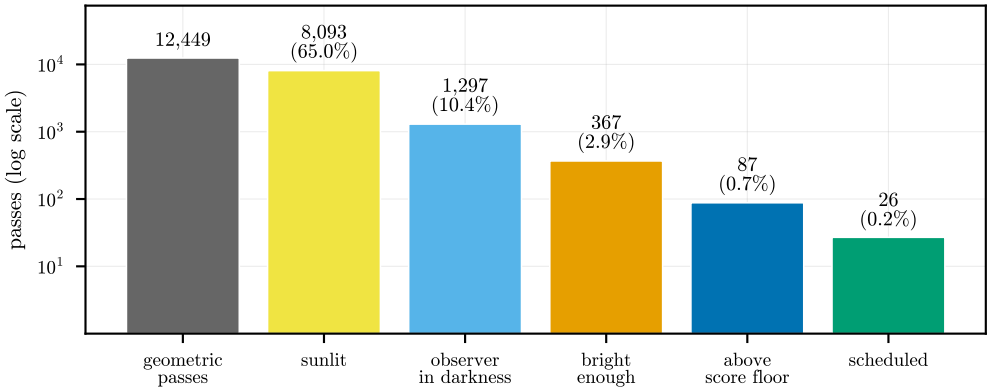


Fig. 3 Visibility funnel for the demonstration station: of 12 449 geometric passes over 7 days, 26 survive to the timetable. Cloud enters the score as a continuous weight rather than a hard cut, so the penultimate stage is the score floor.

Planning 635 objects over 7 days takes 109.4 s single-threaded, dominated by propagation, which scales linearly in objects and horizon. The timetable is stable against the score weights: varying w_e from 0.3 to 0.7 leaves a Jaccard overlap of at least 0.89 with the default, falling to 0.70 once elevation dominates ($w_e \geq 0.9$) and to 0.86 with duration alone; within the moderate range the physical constraints, not the weighting, govern the outcome.

6 Discussion and Limitations

Cloud cover is an imperfect proxy for observing conditions: total cloud fraction on a reanalysis grid cell of order 25 km describes one line of sight only coarsely, and transparency, seeing, cloud height and aerosol are not modelled. A local all-sky camera could supply a better nowcast, to which the calibration of Section 3.3 would apply unchanged, and sub-hourly skill is the route to the one regime in which the present forecast gives no benefit.

The weather-value gains are pooled over six stations and 60 nights in one season. A cluster bootstrap that resamples whole stations (10 000 draws) gives the 95% intervals quoted in Section 5.4; with only six clusters these intervals are approximate and probably too narrow, and the difference between the calibrated and raw planners, whose intervals overlap, was not tested separately.

Both value models are reported because they disagree, and neither captures partial credit; a station with a specific instrument is better served by its own value function, which the scheduler accepts unchanged. One instrument observing one object at a time is assumed, so multiple apertures or a station network would need a different algorithm. Photometry is approximate, (2) treating every object as a diffuse sphere, so the magnitude gate is an ordering rather than a prediction, and the ageing measurement is single-season and limited to low Earth orbit.

7 Conclusion

SkyPass integrates pass prediction, optical visibility analysis, cloud-forecast reasoning and exact conflict resolution into one planner deployable at a small ground station, producing a timetable that exports to a calendar or a rotator controller. Pass extraction agrees with an independent implementation to 0.055 s while requiring 23.7× fewer propagator calls than dense sampling, and conflict resolution is exact at negligible cost.

Two weather findings bear on the design of such systems. First, a forecast is informative about which *nights* merit observation and much less so for ranking passes within a night, where cloud is autocorrelated and pass quality varies more than sky conditions: a planner under a fixed nightly quota gains nothing, whereas given a budget to allocate selectively the same forecast yields 19.0% (expected value) and 37.6% (clear-sky count). Second, calibration against reanalysis raises the gain in every regime tested, and under a quota it is the difference between a loss and break-even. Both findings should carry over to other small facilities allocating scarce effort under uncertain sky, though only one setting is tested here: forecast value is a property of the decision it informs rather than of the forecast alone.

Acknowledgements

The authors thank the Department of Artificial Intelligence and Machine Learning and the Department of Computer Science and Engineering, Saveetha Engineering College, Chennai, for their support of this work, which uses public orbital element sets from CelesTrak and AMSAT and weather data from Open-Meteo.

Statements and Declarations

Funding

No funding was received for conducting this study.

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval and consent

Not applicable: the study involves no human participants, human data or animals.

Data and code availability

SkyPass is open source under the MIT licence at https://github.com/rohithprem18/skypass, with the element sets used and the raw experiment outputs behind every reported value. Weather series are retrieved from Open-Meteo's public archive by the released scripts; historical element sets for the ageing and epoch-correct experiments come from Space-Track, which requires a free account, and are not redistributed.

References

[1] Vallado, D.A.: Fundamentals of Astrodynamics and Applications, 5th edn. Microcosm Press, Hawthorne, CA (2022)

[2] Acciarini, G., Baydin, A.G., Izzo, D.: Closing the gap between SGP4 and high-precision propagation via differentiable programming. Acta Astronautica **226**, 694–701 (2025) https://doi.org/10.1016/j.actaastro.2024.10.063

[3] Li, S., Yu, Q., Ding, H.: Reviews and prospects in satellite range scheduling problem. Autonomous Intelligent Systems **3**(1), 9 (2023) https://doi.org/10.1007/s43684-023-00054-6

[4] Ou, J., *et al.*: Deep reinforcement learning method for satellite range scheduling problem. Swarm and Evolutionary Computation **77**, 101233 (2023) https://doi.org/10.1016/j.swevo.2023.101233

[5] Wang, X., Wu, G., Xing, L., Pedrycz, W.: Agile Earth observation satellite scheduling over 20 years: Formulations, methods, and future directions. IEEE Systems Journal **15**(3), 3881–3892 (2021) https://doi.org/10.1109/jsyst.2020.2997050

[6] Hersbach, H., *et al.*: The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society **146**(730), 1999–2049 (2020) https://doi.org/10.1002/qj.3803

[7] Kovář, P., Puričer, P., Kovářová, K.: Study of the two-line element accuracy by 1U CubeSat with a GPS receiver. Sensors **22**(8), 2902 (2022) https://doi.org/10.3390/s22082902

[8] Fankhauser, F., Tyson, J.A., Askari, J.: Satellite optical brightness. The Astronomical Journal **166**(2), 59 (2023) https://doi.org/10.3847/1538-3881/ace047

[9] Wang, X., Song, G., Leus, R., Han, C.: Robust Earth observation satellite scheduling with uncertainty of cloud coverage. IEEE Transactions on Aerospace and

Electronic Systems **56**(3), 2450–2461 (2020) https://doi.org/10.1109/taes.2019.2947978

[10] Cormen, T.H., Leiserson, C.E., Rivest, R.L., Stein, C.: Introduction to Algorithms, 4th edn. MIT Press, Cambridge, MA (2022)

[11] Xie, H., Kang, Z., Jiang, X.: AstroSA: An astronomical observation scheduler assessment framework in Python. Astronomy and Computing **47**, 100806 (2024) https://doi.org/10.1016/j.ascom.2024.100806

[12] Brown, B., *et al.*: The model evaluation tools (MET): More than a decade of community-supported forecast verification. Bulletin of the American Meteorological Society **102**(4), 782–807 (2021) https://doi.org/10.1175/bams-d-19-0093.1

[13] Vannitsem, S., *et al.*: Statistical postprocessing for weather forecasts: Review, challenges, and avenues in a big data world. Bulletin of the American Meteorological Society **102**(3), 681–699 (2021) https://doi.org/10.1175/bams-d-19-0308.1

[14] Kelso, T.S.: CelesTrak: Current NORAD Two-Line Element Sets. https://celestrak.org/. Accessed 2026-08-29 (2026)

[15] Rhodes, B.: Skyfield: High Precision Research-Grade Positions for Planets and Earth Satellites. Python package, version 1.54, https://rhodesmill.org/skyfield/ (2026)

[16] Zippenfenig, P.: Open-Meteo.com Weather API. https://open-meteo.com/ (2024). https://doi.org/10.5281/zenodo.7970649

[17] Ebert, P.A., Milne, P.: Methodological and conceptual challenges in rare and severe event forecast verification. Natural Hazards and Earth System Sciences **22**(2), 539–557 (2022) https://doi.org/10.5194/nhess-22-539-2022